

.....

***Target ADC input level for 600 mVp-p**
- Gain set to ~4x gain

*** R2 / R14 = 2.2 kΩ**

*** R6 & R14 270ohm - Do not change**

SNR:~70db

THD/SINAD:~98/70

ENOB:~11

Gain table - R2 & R14 Values:

1x 560 - $v_{in}=2.5v_{p-p}$

1.2x 680 - vin=2vp-p

1.7x 910 - vin=1.5vp-p

2.5x 1.2K - vin=1vp-p

3x 1k6 - vin=0.90vp-p

4x 2K2 - $v_{in}=0.65v_{p-p}$

4.1x 2k4 - $v_{in}=0.60v_{p-p}$ ---- Use

5x 3k - Not recommended SNR ~60

Selected for SNR/ENOB and lowest noise

Explanation:

Higher value rf/rg resistors, higher johnson noise.

Analogue Devices AD8138 does not

recommend rf $\geq 5k$, I suggest rf $\geq 3K$

Modification:

=====

- 1. Remove R3,R5,R8,R17**
- 2. Replace R5 with 6.8nF capacitor**
- 3. Replace R9/R11 47ohm**
- 3a. On the AD9226 side and across R9/11 add 180pF**
+Provides 1-pole 10Mhz LPF & ADC kickback suppression
- 4. Replace R2/R14 - 2k2**
- 4a. Add 4.7pF capacitor in parallel with R2/R14**
+Stability and LPF roll off
- 5. R6/R13 Replace, 270ohm**
- 6. Add C3 in series with R13, lift R13 and adding in series**
- 7. Add R16, use two SMD 51ohm Resistors in parallel**
+Required for DC offset balance

